

Gm/ID Project Instructions

Learning Objectives:

By the end of this project, you should be able to:

- **LO1:** Describe the electrical behaviour of the MOS transistor in scaled technologies, taking into consideration short-channel effects.
- **LO2:** Design the region of inversion (weak, moderate, strong) and level of saturation (triode, saturation) of each transistor in your circuit to meet the requirements of your application.
- **LO3:** Size optimally all the transistors in any advanced technology node using the gm/ID methodology.

Group Work: You can work in groups of a **maximum of 2** people.

Project Description:

Your boss saw the design example in the gm/ID lecture. She says that this design is close to what your company needs for a product but wants to enlist your skills for an improved version that meets the following objectives:

- Design an intrinsic gain stage (IGS) with capacitive feedback. But, instead of biasing the NMOS (M1) using a constant current source, you are asked to replace it with a PMOS (M2). You can bias that PMOS using a constant voltage source or a diode-connected PMOS (M3) sourced by a constant current source (Fig.1).
- The total **Area must be minimized** as much as possible given a drain current budget of **500 μ A**. Note: The area to be minimized in the task is only the area of M1 + M2.
- The **settling time** must be below **5.5 ns** with self-loading included.
- The **static error** needs to be below **8%**.
- Optimize the ratio **gm_p/gm_n** to maximize your **dynamic range (DR)** $DR = \frac{P_{out}}{P_{n,out}}$ (assume sinusoidal signal for P_{out}).
- All other parameters (**FO, G, etc.**) remain the same.

Table 1: Required Specifications Summary

Specification	Value
Static Error [%]	≤ 8
Settling time [ns] (@ 0.1% ϵ_d)	≤ 5.5
IDmax [μ A]	500
Total integrated output noise [μ Vrms] (10:10G)Hz	≤ 100
Close loop gain (G)	2
Fan out (FO)	1

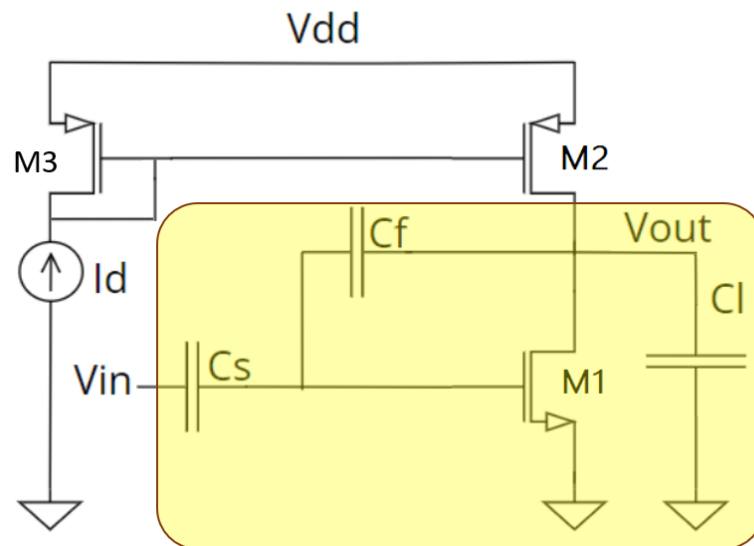


Figure 1: Schematic of the IGS with an active PMOS load and capacitive feedback network.

Deliverables:

Submit the following materials on **Brightspace**:

1. **A PDF Report (Maximum 4 pages, IEEE Transactions format).** The report must include:
 - A **schematic** of your final design.
 - A **short description** of your optimization approach, including a **flowchart** to describe the MATLAB code and final design parameters.
 - A **comparison discussion** between MATLAB computed results and Spice results (**settling time, static error, total integrated noise**). The discrepancy should be ~10% or less.
 - **Plots** for:
 - 1- **Settling transient** (annotate the settling value).
 - 2- **Dynamic Error** (annotate the Settling time).
 - 3- **Running output noise integral** (versus frequency), annotated with the final μVrms value.
 - 4- Any other plot used in the optimization problem.
 2. **MATLAB Code:** The MATLAB script used for your project. Ensure that it is **self-contained and error-free**, as it will be run for grading. Otherwise, grading will not be possible.
 3. **LTSpice Files:** All necessary LTSpice simulation files. Ensure they run properly before submission.
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Tables to be included:

- 1- Comparison of the obtained results using the Matlab script with the simulated using Spice.

Parameter	Matlab Value	Spice sim. Value	Error (%)
Wn [μm]			
Ln [μm]			
Wp [μm]			
Lp [μm]			
Total Area [μm^2]			
CR			
DR			
(gm/ID) _n [S/A]			
(gm/ID) _p [S/A]			
Total integrated output Noise [μV_{rms}]			
Settling Time [ns]			
Static Error [%]			
ID [μA]			

Grading Rubric

Design Specifications (settling time, static error, noise, etc.) **20%**

- All requirements met **Area is minimized** and **DR is maximized**

MATLAB Script – with all the required loops as gm/ID loop, sizing loop, etc. **40%**

- Correct implementation **without errors**
- Flow chart

LTSpice Simulations **10%**

- Fine-tuned for best-in-class specs; all simulations correctly set

Design Methodology **30%**

- Design choices are well-motivated (e.g., **gm/ID level**)
- Shows a clear understanding of MOS physics
- Discusses MATLAB vs. LTSpice discrepancies thoroughly.

Final Notes:

- You can modify the starter file "**igs_cap_sizing.m**" as needed (e.g., add for-loops, change design flow, etc.).
- Ensure all files are correctly formatted and meet the specified requirements.
- It is very hard to get the same results for two different groups. So avoid intergroup working.
- Use **systematic design approaches** instead of brute-force tweaking.
- Validate all results and minimize discrepancies.
- Good luck with your design!